

Tsung-Han Yang

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EDUCATION	Ph.D. Physics Brown University, Providence, RI, USA	2023
	M.Sc. Physics National Taiwan University, Taipei, Taiwan	2012
	B.S. Physics National Chung Hsing University, Taichung, Taiwan	2010
RESEARCH EXPERIENCE	Postdoctoral Research Fellow Oak Ridge National Laboratory, Spallation Neutron Source, Oak Ridge, TN	2023 - Present
	• Independent Research Leadership: Led the investigation of topological phases and short-range correlations in Kagome magnets, rare-earth compounds, and high-entropy materials; successful execution of these projects resulted in first-author publications in <i>Nat. Commun.</i> and <i>JACS</i>. • Multi-Modal Instrumentation Expertise: Design and execute complex neutron scattering experiments to characterize quantum material properties, utilizing a diverse portfolio of SNS and HFIR instruments: – <i>Diffraction:</i> POWGEN, TOPAZ, DEMAND – <i>Total and Diffuse scattering:</i> NOMAD, CORELLI – <i>Inelastic Spectroscopy:</i> ARCS, CNCS, SEQUOIA • Advanced Computational Modeling: Implement specialized analysis techniques, including Reverse Monte Carlo (RMC) modeling, Pair Distribution Function (PDF) analysis, magnetic PDF, diffuse scattering analysis, and refinements to resolve local/average atomic distortions and magnetic structures. • Mentorship & Collaboration: Provide technical guidance and hands-on training to visiting graduate students and research groups at POWGEN, overseeing experimental setup and data acquisition strategies. • Peer Review & Scientific Service: Selected as an invited reviewer for <i>Nat. Commun.</i>, providing expert evaluation of manuscripts on quantum materials with emphasis on symmetry analysis and experimental validation.	
	Graduate Research Assistant Brown University, Plumb Lab, Providence, RI	2018 - 2023
	• Laboratory Commissioning (Infrastructure): Designed and built a dedicated High-Pressure X-ray Diffraction laboratory from the ground up, integrating Diamond Anvil Cells (DAC) with low-temperature cryostats for in-house characterization. • Competitive Beamtime Leadership: Secured resources as Principal Investigator through successful peer-reviewed proposals at major national facilities, including SNS, HFIR, APS, NSLS-II, and CHESS. • Synchrotron & Neutron Experimentation: Executed extensive experiments across multiple national user facilities, utilizing complementary probes:	

- *Synchrotron X-ray (APS, NSLS-II, CHESS)*: High-resolution inelastic X-ray scattering (HERIX), RIXS, X-ray Absorption Spectroscopy (XAS), Crystallography, and Total scattering (PDF).
- *Neutron Scattering (SNS, HFIR)*: Time-of-flight spectroscopy (ARCS, CNCS, SEQUOIA) and total scattering (NOMAD).
- **Material Synthesis**: Synthesized high-quality single crystals and polycrystals (Lacunar spinels) to support the experimental program.
- **Computational Analysis**: Developed custom Python analysis pipelines, performed Density Functional Theory (DFT) and phonon calculations to corroborate experimental findings.

Teaching Assistant

2015 - 2018

[Brown University](#), Providence, RI

- **Laboratory Instruction**: Instructed undergraduate students in "Fundamental Physics" (approx. 40 students per semester), delivering pre-lab lectures on experimental theory and enforcing strict laboratory safety protocols.
- **Student Assessment & Mentorship**: Evaluated weekly laboratory reports with a focus on data analysis, error propagation, and scientific writing.

Research Assistant

2013 - 2015

[Academia Sinica](#), [Research Center of Applied Science](#), Taipei, Taiwan

- **Ab Initio Electronic Structure**: Employed Density Functional Theory (DFT) to investigate the electronic properties and molecule-substrate coupling in organic spintronic systems.
- **Quantum Transport Simulation**: Modeled spin-dependent transport in metal-organic hybrid systems using the Non-Equilibrium Green's Function (NEGF) formalism combined with DFT.
- **Theory-Experiment Synergy**: Collaborated closely with Scanning Tunneling Microscopy (STM) groups to interpret experimental spectra, directly correlating theoretical density of states with experimental dI/dV measurements.

Graduate Research Assistant

2010 - 2012

[National Taiwan University](#), Taipei, Taiwan

- **Thin Film Fabrication**: Synthesized epitaxial thin film materials using Molecular Beam Epitaxy (MBE) under ultra-high vacuum conditions.
- **Surface Characterization (STM)**: Characterized atomic-scale surface magnetic properties using Spin-Polarized Scanning Tunneling Microscopy (SP-STM) in an Ultra-High Vacuum (UHV) environment at liquid helium temperatures (4 K).
- **Theoretical Correlation**: Performed Density Functional Theory (DFT) calculations to corroborate experimental observations of surface electronic structures.

TECHNICAL SKILLS

- **Scattering Techniques**: Inelastic Neutron Scattering (Time-of-Flight), Neutron/X-ray Powder/Single Crystal Diffraction, Total Scattering (PDF), Resonant Inelastic X-ray Scattering (RIXS), High-Resolution Inelastic X-ray Scattering (HERIX), X-ray Absorption Spectroscopy (XAS).
- **Laboratory Instrumentation**: High-Pressure Diamond Anvil Cells (DAC), Cryogenics (Dilution/He3/LHe), Crystal Growth (Flux/CVT methods), Glove-box, Molecular Beam Epitaxy (MBE), Spin-Polarized STM (UHV/Low-T).

- **Computation & Analysis:** Density Functional Theory (DFT), Ab Initio Molecular Dynamics (AIMD), Frozen Phonon Calculations, Reverse Monte Carlo (RMC), Rietveld Refinement, Magnetic Symmetry Analysis.
- **Programming & Software:** Python (SciPy, NumPy, Matplotlib), Fortran, Bash, Mantid, VASP/Quantum ESPRESSO, Jana2020, GSAS-II, FullProf.

PUBLICATIONS

7. **T.-H. Yang**, S. Okamoto, D. A. Tennant, M. A. McGuire and Q. Zhang. “Intrinsic Topological Weyl Phase Transition Induced by a Magnetostructural Transformation in a Kagome Magnet.” *Nat. Commun.* (2026).
6. **T.-H. Yang**, G. Shan, Y. Zhang, D. Olds, W. Meier, M. B. Stone, B. C. Sales, A. D. Christianson and Q. Zhang. “Simultaneous Development of Antiferromagnetism and Local Symmetry Breaking in a Kagome Magnet ($\text{Fe}_{0.55}\text{Co}_{0.45}\text{Sn}$).” *J. Am. Chem. Soc.* **146**, 50, 34374–34382 (2024).
5. **T.-H. Yang**, T. Chang, Y.-S. Chen, and K. W. Plumb. “Jahn-Teller driven quadrupolar ordering and spin-orbital dimer formation in GaNb_4Se_8 .” *Phys. Rev. B* **109** (14), 144101 (2024).
4. **T.-H. Yang**, S. Kawamoto, T. Higo, S. G. Wang, M. B. Stone, J. Neuefeind, J. P. C. Ruff, A. M. M. Abeykoon, Y.-S. Chen, S. Nakatsuji, and K. W. Plumb. “Bond Ordering and Molecular Spin-orbital Fluctuations in the Cluster Mott Insulator GaTa_4Se_8 .” *Phys. Rev. Research* **4**, 033123 (2022).
3. H.-H. Yang, H.-H. Tsai, C.-F. Ying, **T.-H. Yang**, C.-C. Kaun, C.-H. Chen, and M.-T. Lin. “Tuning Molecule-Substrate Coupling via Deposition of Metal Adatoms.” *J. Chem. Phys.* **143**, 184704 (2015).
2. Y.-H. Chu, C.-H. Hsu, C.-I. Lu, H.-H. Yang, **T.-H. Yang**, C.-H. Luo, K.-J. Yang, S.-H. Hsu, C.-C. Kaun, G. Hoffmann, and M.-T. Lin. “Spin-Dependent Molecule Symmetry at a Pentacene-Co Spinterface.” *ACS Nano* **9**, 7027 (2015).
1. H.-H. Yang, Y.-H. Chu, C.-I. Lu, **T.-H. Yang**, K.-J. Yang, C.-C. Kaun, G. Hoffmann, and M.-T. Lin. “Digitized Charge Transfer Magnitude Determined by Metal-Organic Coordination Number.” *ACS Nano* **7**, 2814 (2013).

FIRST AUTHORED MANUSCRIPT

2. **T.-H. Yang**, S. H. Lee, H. Tan, Y. Zhang, D. Olds, V. Petříček, M. G. Tucker, H. Cao, J. Yan, B. Yan, Z. Mao, and Q. Zhang. “Local Symmetry Breaking-Activated Anomalous Hall Effect in the Kagome Magnet Mn_3Sn .” *Manuscript in preparation*.
1. **T.-H. Yang**, D. L. Abernathy, G. Kwon, A. M. M. Abeykoon, E. Bozin, and K. W. Plumb. “Spin-orbital fluctuation in cluster Mott insulator GaM_4X_8 .” *Manuscript in preparation*.

DISSERTATIONS

2. T.-H. Yang, “Spin-orbital Fluctuations and Unconventional Quantum Phases in Lacunar Spinels: Insights from Neutron and X-ray Scattering”, Brown University (2023).
1. T.-H. Yang, “Investigating Electronic Structure and Magnetic Coupling in Metal-organic Hybrid System”, National Taiwan University (2012).

INVITED PRESENTATIONS

3. “Simultaneous Local Structural and Global Magnetic Transition in a Kagome Magnet,” **American Crystallographic Association (ACA) Annual Meeting**, Lombard, OH (2025).
2. Neutron Scattering Division Representative: “Department of Energy (DOE) Research Highlight,” Representing the Neutron Science Division (ORNL), **U.S. Department of Energy**, Virtual (2025).
1. “Bond Ordering in Cluster Mott Insulator GaTa_4Se_8 ,” Materials Science Division Seminar, **Argonne National Laboratory**, Lemont, IL (2023).

AWARDS

- Poster Award, Annual Meeting of American Neutron Scattering Society, USA 2022.
- Dean’s Award of College of Science, National Taiwan University 2012.
- Best Master’s Thesis Award, Taiwan Association for Magnetic Technology, Taiwan 2012.
- Poster Award, Annual Meeting of the Physical Society of Taiwan, Taiwan 2011.
- Outstanding Incoming Graduate Student Scholarship, National Taiwan University 2010.